Experiment 7

Random Forest

Aim

To implement the Random Forest algorithm as an ensemble of multiple decision trees using random sampling and feature selection, and to evaluate its performance for classification tasks.

Theory

A Random Forest is a supervised ensemble learning algorithm used for both classification and regression tasks. Instead of relying on a single decision tree (which can easily overfit the training data), Random Forest builds many decision trees during training and combines their outputs to produce a more accurate and stable prediction. Each individual tree in the forest may be a relatively weak learner on its own, but when many such trees vote together, the ensemble becomes a strong learner with significantly improved generalisation.

1. From a Single Decision Tree to a Forest

A single Decision Tree learns by recursively splitting data on features to create decision rules. While simple and interpretable, it has a key limitation: high variance. Small changes in training data can produce a completely different tree, leading to overfitting. Random Forest addresses this by building an ensemble of many decision trees and aggregating their predictions. The core idea is:

Many weak learners, each slightly different, can combine to form one strong learner.

The two sources of randomness that make the trees diverse are Bootstrap Sampling (Bagging) and Random Feature Selection, explained below.

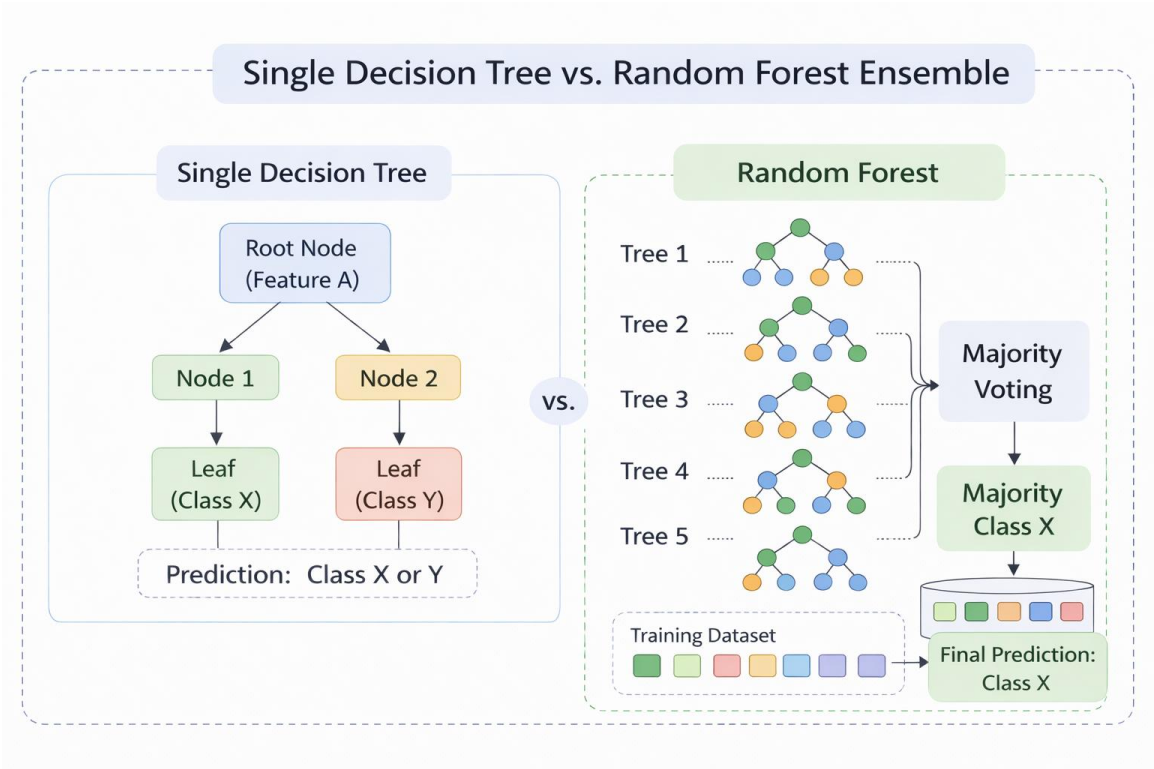


Figure 1 - A single Decision Tree (left) vs. a Random Forest ensemble (right). Each tree votes independently and the final prediction is made via majority voting.

2. Bootstrap Sampling (Bagging)

Bagging (Bootstrap Aggregating) is the first source of randomness. For each tree in the forest:

1. A bootstrap sample is created by randomly selecting N samples with replacement from the original training set of N samples.
2. On average, each bootstrap sample contains about 63% of unique training samples (since sampling is with replacement, some samples appear more than once and some are left out).
3. The remaining ~37% of samples that are not selected for a given tree are called Out-of-Bag (OOB) samples. These act as a built-in validation set for that tree.

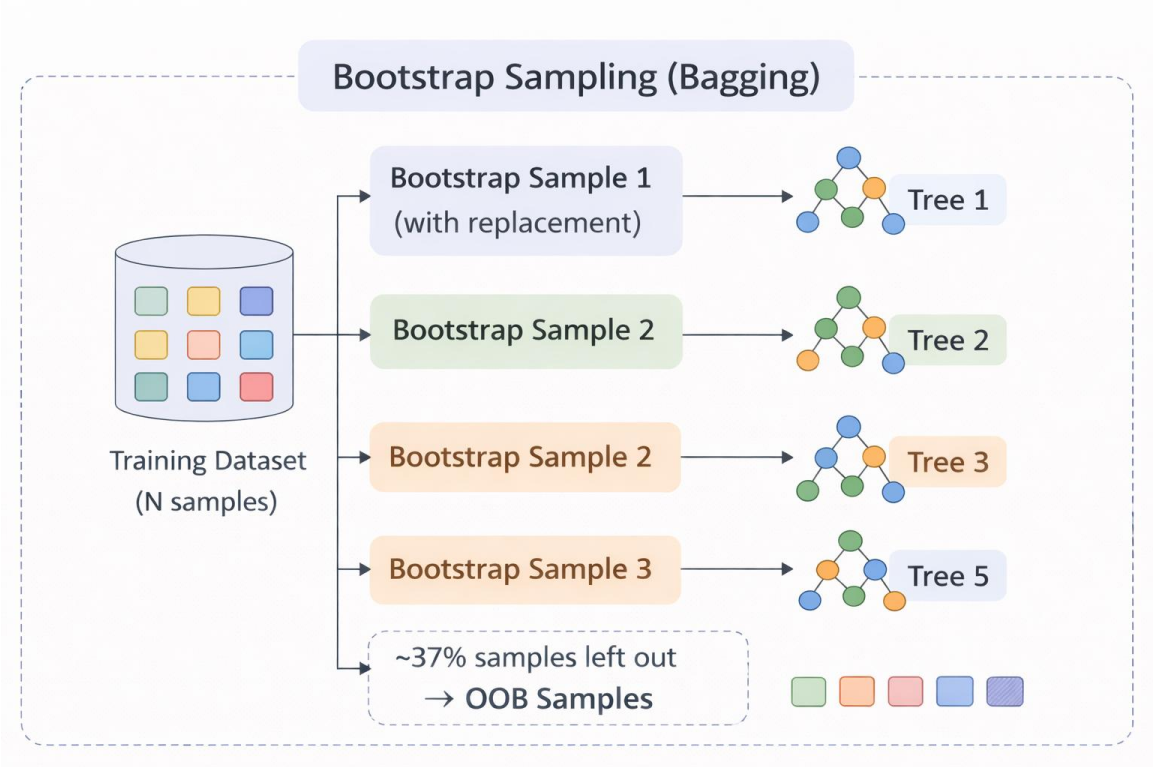


Figure 2 - Bootstrap sampling: each tree receives a different random subset (with replacement) of the training data. Unselected samples become OOB samples.

Because each tree is trained on a different subset, the trees learn slightly different patterns, making their errors less correlated. When combined, the variance of the overall model drops significantly.

3. Random Feature Selection

The second source of randomness occurs at every split during tree construction. Instead of evaluating all available features to find the best split, each node only considers a random subset of m features (typically $m = \sqrt{p}$ for classification, where p is the total number of features).

This prevents dominant features from appearing at the root of every tree, forcing the forest to explore a wider variety of feature interactions. Without this step, all trees would look very similar despite using different bootstrap samples.

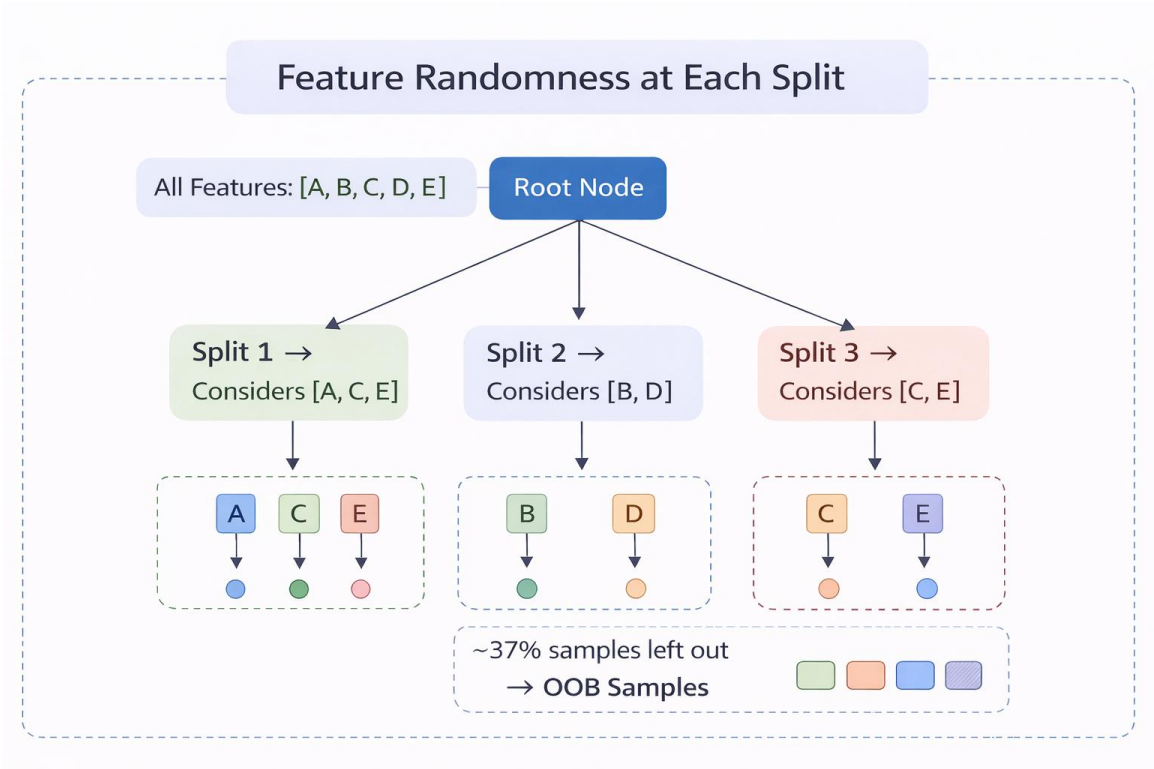


Figure 3 - At each node, only a random subset of features (highlighted) is evaluated. Different nodes and different trees use different subsets.

4. Bagging + Feature Randomness: Working Together

The power of Random Forest comes from combining both sources of randomness:

Source of Randomness	What it does	Effect
Bootstrap Sampling	Each tree trains on a different data subset	Reduces variance by decorrelating trees
Feature Randomness	Each split considers a random feature subset	Prevents dominant features from making trees similar
Combined Effect	Trees are diverse in both data and features	Ensemble is robust, low-variance, and generalises well

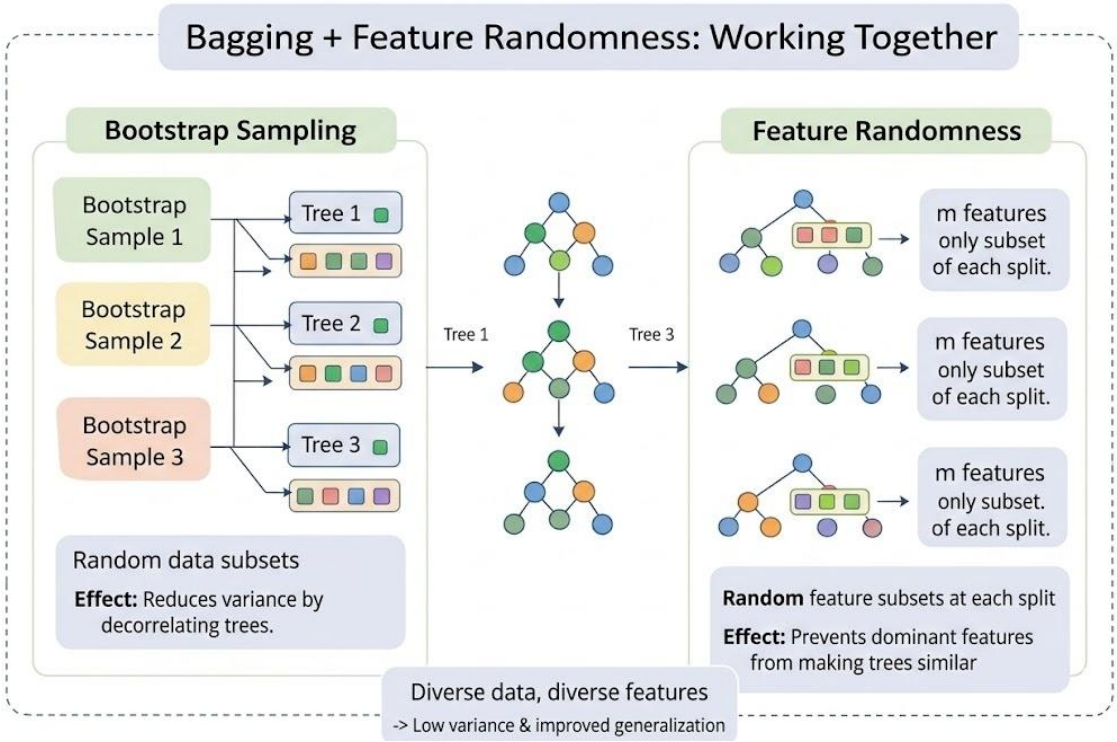


Figure 4 - Random Forest = Bagging (different data subsets) + Feature Randomness (different feature subsets at each split). This dual randomness produces a diverse, high-performing ensemble.

5. Splitting Criteria

At each node, the best split is chosen from the randomly selected feature subset using an impurity measure. The two commonly used criteria are:

Gini Impurity measures the probability of misclassifying a randomly chosen sample:

$$Gini = 1 - \sum_{i=1}^{C} p_i^2$$

where p_i is the proportion of samples belonging to class i, and C is the total number of classes. A Gini value of 0 indicates a perfectly pure node (all samples belong to one class). Higher values indicate more impurity.

Entropy measures the amount of uncertainty or disorder in a node:

$$Entropy = -\sum_{i=1}^{C} p_i \, log_2(p_i)$$

Entropy is 0 for a pure node and reaches its maximum when classes are equally distributed. Information Gain, which is the reduction in entropy after a split, is used to select the best feature and threshold.

Both criteria generally produce similar trees. Gini is computationally slightly faster (no logarithm), while Entropy tends to produce more balanced splits.

6. Prediction Mechanism

Once all trees are trained, a new sample is passed through every tree in the forest:

- Classification: Each tree votes for a class. The class with the majority of votes becomes the final prediction.
- Regression: Each tree outputs a numeric value. The average of all outputs is the final prediction.

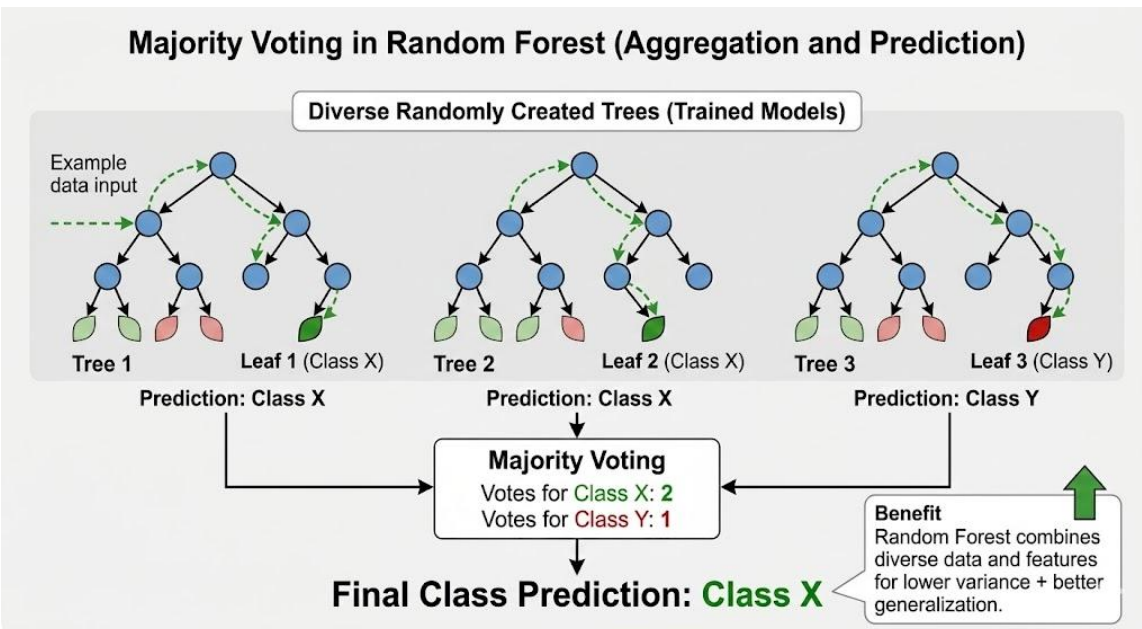


Figure 5 - Prediction via majority voting: each tree votes independently, and the class with the most votes wins.

Example

Suppose a Random Forest contains 5 trees and predicts whether an email is Spam or Not Spam:

Tree	Prediction
Tree 1	Spam
Tree 2	Not Spam
Tree 3	Spam
Tree 4	Spam
Tree 5	Not Spam

Majority vote = Spam (3 out of 5 trees). Even though two trees predicted "Not Spam", the ensemble correctly classifies the email by following the majority.

7. Out-of-Bag (OOB) Error Estimation

A unique advantage of Random Forest is built-in cross-validation through OOB samples. Since each tree was trained on only ~63% of the data, the remaining ~37% (OOB samples) can be used to evaluate that tree's performance without needing a separate validation set.

The OOB error is computed by:

4. For each training sample, collecting predictions only from trees where it was not in the bootstrap sample.
5. Using majority voting on those predictions to classify the sample.
6. Computing the overall misclassification rate.

The OOB error is an unbiased estimate of the generalisation error and closely approximates leave-one-out cross-validation.

8. Key Hyperparameters

Parameter	Description	Typical Values
n_estimators	Number of trees in the forest	100–500 (more trees → better accuracy, diminishing returns)
max_features	Number of features considered at each split	√p (classification), p/3 (regression)
max_depth	Maximum depth of each tree	None (grow fully) or limited to prevent overfitting
min_samples_split	Minimum samples required to split a node	2–10
min_samples_leaf	Minimum samples required at a leaf node	1–5

Increasing n_estimators generally improves performance but increases computation time. Reducing max_features increases tree diversity but may reduce individual tree accuracy. The balance between these parameters is key to a well-performing Random Forest.

9. Algorithm

- Step 1: Set parameters:
  n = number of trees to build
  m = number of features to consider at each split (usually √total_features for classification)

- Step 2: For each tree i = 1 to n:
  - Step 2a: Bootstrap Sampling

Randomly select N samples with replacement from training data
This creates a different dataset for each tree
About 37% of original data is left out (Out-of-Bag samples)

- Step 2b: Build Decision Tree with Random Feature Selection
  At each node:
  Randomly select m features from all features
  Find the best split using ONLY these m features
  Split the node
  Grow tree fully (no pruning)

- Step 3: Store all n trees

- Step 4: For prediction (Classification):
  Pass new sample through all trees
  Each tree gives one vote
  Final prediction = class with majority votes

- Step 5: For prediction (Regression):
  Pass new sample through all trees
  Each tree gives one numeric prediction
  Final prediction = average of all tree predictions

- Step 6: Evaluate using OOB Error:
  For each sample, aggregate predictions from trees that did not include it
  Compute misclassification rate as OOB error estimate

- Step 7: Calculate Feature Importance:
  For each feature, sum up how much it reduced impurity across all trees

10. Decision Tree vs. Random Forest

Aspect	**Single Decision Tree**	**Random Forest**
Model Type	Single model	Ensemble of many trees
Variance	High (sensitive to data changes)	Low (averaging reduces variance)
Overfitting	Prone to overfitting	Resistant due to bagging + randomness
Accuracy	Moderate	Generally higher
Interpretability	Easy (one tree to follow)	Harder (many trees)
Training Time	Fast	Slower (many trees)
Feature Importance	Available but unstable	Stable and reliable

11. Merits of Random Forest

- High Accuracy & Robustness: Combining multiple trees reduces overfitting and usually gives better accuracy than a single decision tree.
- Handles Large & Complex Data: Works well with large datasets and can handle both numerical and categorical features.
- Feature Importance Estimation: Measures the importance of each feature based on impurity reduction, aiding feature selection.
- Built-in Validation: OOB error provides a reliable generalisation estimate without needing a separate validation set.
- Minimal Preprocessing: Does not require feature scaling or normalisation.

12. Demerits of Random Forest

- Computationally Expensive: Training many trees requires more time and memory compared to simpler models.
- Less Interpretability: The ensemble nature makes it harder to interpret compared to a single decision tree.
- Not Ideal for Real-Time Prediction: Multiple trees increase prediction latency.
- Large Model Size: Storing hundreds of trees consumes more memory than a single model.

Random Forest - Practical Implementation

Option A: Binary Classification (Bank Marketing Dataset)

Step 1: Importing Libraries

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

print("Libraries imported successfully")
```

Output:

```
Libraries imported successfully
```

Step 2: Loading Dataset

Import necessary libraries

```
df = pd.read_csv("bank_marketing.csv", sep=';')

print("Dataset loaded successfully")
```

Output:

```
Dataset loaded successfully
```

Step 3: Data Analysis

Check dataset dimensions

```
print("Dataset shape:", df.shape)
```

Output:

```
Dataset shape: (41188, 21)
```

Preview first 5 records of the dataset

```
print("First 5 rows:")
print(df.head())
```

Output:

```
First 5 rows:
```

Output:

	age	job	marital	education	default	housing	loan	contact	...
0	56	housemaid	married	basic.4y	no	no	no	telephone	...
1	57	services	married	high.school	unknown	no	no	telephone	...
2	37	services	married	high.school	no	yes	no	telephone	...
3	40	admin.	married	basic.6y	no	no	no	telephone	...
4	56	services	married	high.school	no	no	yes	telephone	...

Output:

```
[5 rows x 21 columns]
```

Check dataset information and types

```
print("Dataset information:")
df.info()
```

Output:

```
Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 41188 entries, 0 to 41187
Data columns (total 21 columns):
#   Column          Non-Null Count  Dtype
---  ------          --------------  -----
0   age             41188 non-null  int64
1   job             41188 non-null  object
2   marital         41188 non-null  object
3   education       41188 non-null  object
4   default         41188 non-null  object
5   housing         41188 non-null  object
6   loan            41188 non-null  object
7   contact         41188 non-null  object
8   month           41188 non-null  object
9   day_of_week     41188 non-null  object
10  duration        41188 non-null  int64
11  campaign        41188 non-null  int64
12  pdays           41188 non-null  int64
13  previous        41188 non-null  int64
14  poutcome        41188 non-null  object
15  emp.var.rate    41188 non-null  float64
16  cons.price.idx  41188 non-null  float64
17  cons.conf.idx   41188 non-null  float64
18  euribor3m       41188 non-null  float64
19  nr.employed     41188 non-null  float64
20  y               41188 non-null  object
dtypes: float64(5), int64(5), object(11)
memory usage: 6.6+ MB
```

Step 4: Data Preprocessing

Perform Label Encoding on categorical variables

```
df_encoded = df.copy()
le = LabelEncoder()

for col in df_encoded.columns:
    if df_encoded[col].dtype == 'object':
        df_encoded[col] = le.fit_transform(df_encoded[col])
        print(f"Encoded column: {col}")

print("\\nCategorical encoding completed")
```

Output:

```
Encoded column: job
Encoded column: marital
Encoded column: education
Encoded column: default
Encoded column: housing
Encoded column: loan
Encoded column: contact
Encoded column: month
Encoded column: day_of_week
Encoded column: poutcome
Encoded column: y
Categorical encoding completed
```

Separate independent and dependent variables

```
X = df_encoded.drop('y', axis=1)
y = df_encoded['y']

print("Features and target separated")
```

Output:

```
Features and target separated
```

Split data into Training and Testing sets

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print(f"Training set: {len(X_train)} samples")
print(f"Testing set: {len(X_test)} samples")
```

Output:

```
Training set: 32950 samples
Testing set: 8238 samples
```

Option A-i: Binary - Gini Impurity

Step 5: Model Training

Initialize Random Forest classifier with Gini criterion

```
rf_model = RandomForestClassifier(
    n_estimators=200,
    max_depth=15,
    min_samples_split=5,
    min_samples_leaf=2,
    random_state=42,
    oob_score=True,
    n_jobs=-1
)

print("Random Forest model initialized with Gini criterion")
print("OOB scoring enabled for generalization estimate")
```

Output:

```
Random Forest model initialized with Gini criterion
OOB scoring enabled for generalization estimate
```

Fit the model on training data

```
rf_model.fit(X_train, y_train)

print("Model training completed")
print(rf_model)
```

Output:

```
Model training completed
RandomForestClassifier(max_depth=15, min_samples_leaf=2, min_samples_split=5, n_estimators=200,
n_jobs=-1, oob_score=True, random_state=42)
```

Step 6: Model Evaluation

Generate predictions on test data

```
y_pred = rf_model.predict(X_test)

print(f"Predictions generated for {len(y_pred)} test samples")
```

Output:

```
Predictions generated for 8238 test samples
```

Calculate overall accuracy of the model

```
accuracy = accuracy_score(y_test, y_pred)

print(f"Accuracy of Random Forest model: {accuracy}")
```

Output:

```
Accuracy of Random Forest model: 0.9225540179655256
```

Out-of-Bag (OOB) Score -- each tree tested on data it never saw during bootstrap

```
print(f"OOB Score: {rf_model.oob_score_:.4f}")
print()
print("The OOB score estimates generalization accuracy without")
print("needing a separate validation set. Each tree is tested only")
print("on the ~37% of samples excluded from its bootstrap sample.")
```

Output:

```
OOB Score: 0.9160
The OOB score estimates generalization accuracy without
needing a separate validation set. Each tree is tested only
on the ~37% of samples excluded from its bootstrap sample.
```

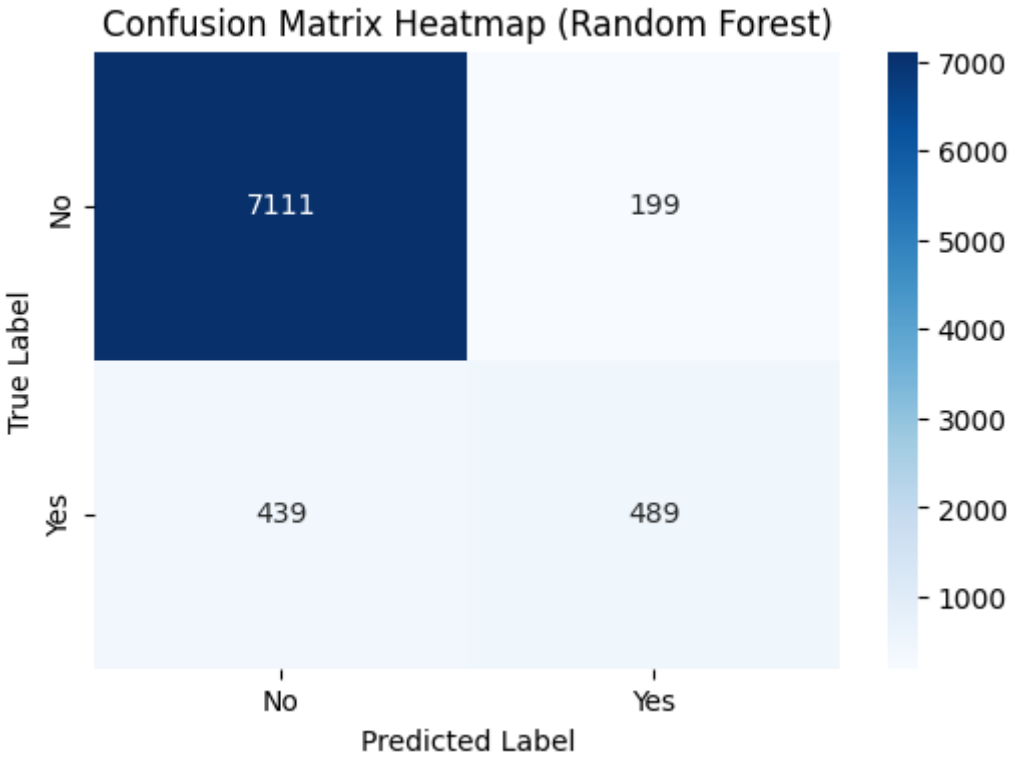
Display detailed classification report

```
print(classification_report(y_test, y_pred))
```

Output:

```
Classification Report:
precision    recall  f1-score   support
0       0.94      0.97      0.96      7310
1       0.71      0.53      0.61       928
accuracy                           0.92      8238
macro avg       0.83      0.75      0.78      8238
weighted avg       0.92      0.92      0.92      8238
```

```
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6, 4))
sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues',
    xticklabels=['No', 'Yes'],
    yticklabels=['No', 'Yes']
)
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.title("Confusion Matrix Heatmap (Random Forest)")
plt.show()
```



Step 7: Model Simulation

Test prediction on sample data

```
sample_data = [[56, 1, 1, 0, 0, 0, 0, 1, 0, 0, 261, 1, 999, 0, 0, 1.1, 93.994, -36.4,
    4.857, 5191.0]]

prediction = rf_model.predict(sample_data)

if prediction[0] == 0:
    print("Prediction: No (Customer will NOT subscribe)")
else:
    print("Prediction: Yes (Customer will subscribe)")
```

Output:

```
Prediction: No (Customer will NOT subscribe)
```

Option A-ii: Binary - Entropy

Step 5: Model Training

Initialize Random Forest classifier with Entropy criterion

```
rf_entropy = RandomForestClassifier(
    criterion='entropy',
    random_state=42,
    n_estimators=200,
    max_depth=15,
    min_samples_split=5,
    min_samples_leaf=2,
    oob_score=True,
    n_jobs=-1
)

print("Random Forest model initialized with Entropy criterion")
print("OOB scoring enabled for generalization estimate")
```

Output:

```
Random Forest model initialized with Entropy criterion
OOB scoring enabled for generalization estimate
```

Fit the model on training data

```
rf_entropy.fit(X_train, y_train)

print("Model training completed")
```

Output:

```
Model training completed
```

Step 6: Model Evaluation

Generate predictions on test data

```
y_pred = rf_entropy.predict(X_test)

print(f"Predictions generated for {len(y_pred)} test samples")
```

Output:

```
Predictions generated for 8238 test samples
```

Calculate overall model accuracy

```
accuracy = accuracy_score(y_test, y_pred)

print(f"Overall Accuracy: {accuracy:.4f}")
```

Output:

```
Overall Accuracy: 0.9226
```

Out-of-Bag (OOB) Score -- each tree tested on data it never saw during bootstrap

```
print(f"OOB Score: {rf_entropy.oob_score_:.4f}")
print()
print("The OOB score estimates generalization accuracy without")
print("needing a separate validation set. Each tree is tested only")
print("on the ~37% of samples excluded from its bootstrap sample.")
```

Output:

```
OOB Score: 0.9148
The OOB score estimates generalization accuracy without
needing a separate validation set. Each tree is tested only
on the ~37% of samples excluded from its bootstrap sample.
```

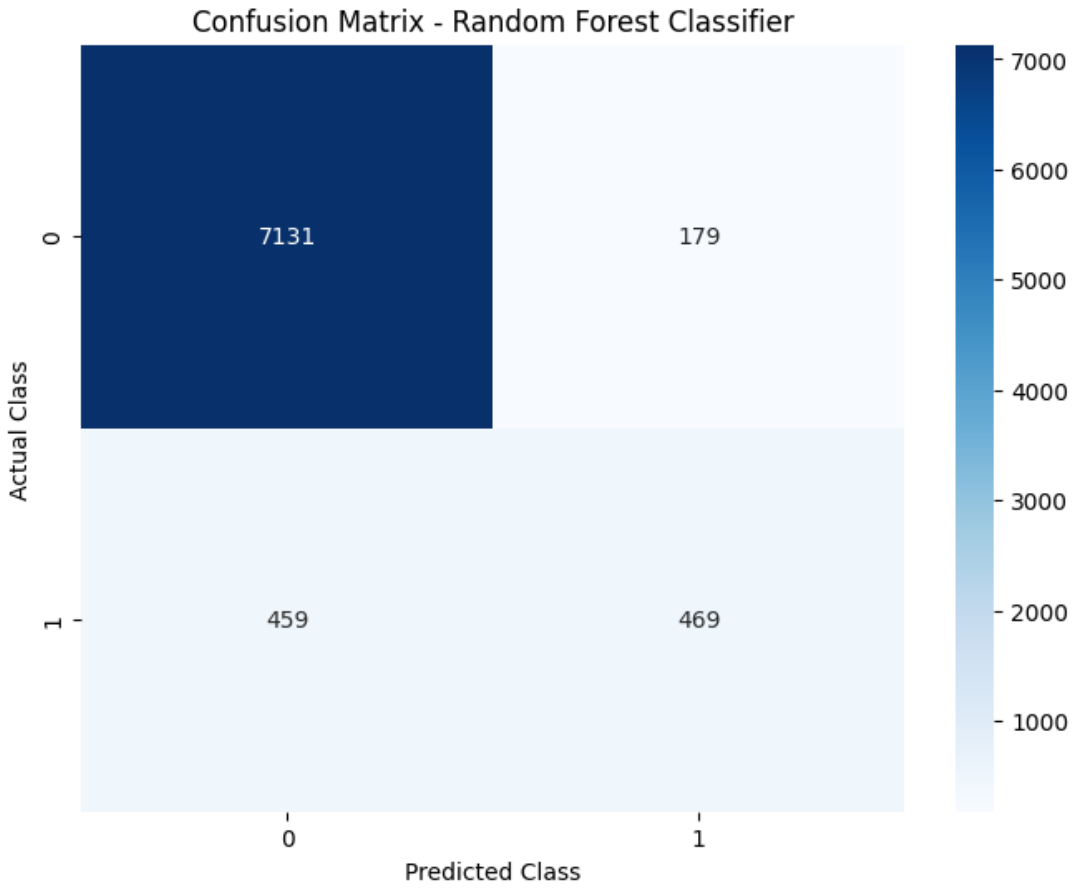
Display detailed classification report

```
print(classification_report(y_test, y_pred))
```

Output:

```
Classification Report:
precision    recall  f1-score   support
0       0.94      0.98      0.96      7310
1       0.72      0.51      0.60       928
accuracy                           0.92      8238
macro avg       0.83      0.74      0.78      8238
weighted avg       0.92      0.92      0.92      8238
```

```
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(8, 6))
sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues',
    xticklabels=['No', 'Yes'],
    yticklabels=['No', 'Yes']
)
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.title("Confusion Matrix - Random Forest Classifier")
plt.show()
```



Step 7: Model Simulation

Test prediction on sample data

```
sample_data = [[56, 1, 1, 0, 0, 0, 0, 1, 0, 0, 261, 1, 999, 0, 0, 1.1, 93.994, -36.4,
    4.857, 5191.0]]

prediction = rf_entropy.predict(sample_data)

if prediction[0] == 0:
    print("Prediction: No (Customer will NOT subscribe)")
else:
    print("Prediction: Yes (Customer will subscribe)")
```

Output:

```
Prediction: No (Customer will NOT subscribe)
```

Option A-iii: Binary - Both (Gini vs Entropy)

Step 5: Model Training

Initialize Random Forest with Gini impurity criterion

```
rf_gini = RandomForestClassifier(
    criterion='gini',
    n_estimators=200,
    max_depth=15,
    min_samples_split=5,
    min_samples_leaf=2,
    oob_score=True,
    random_state=42,
    n_jobs=-1
)

print("Gini model initialized")
print(rf_gini)
```

Output:

```
Gini model initialized
RandomForestClassifier(criterion='gini', max_depth=15, min_samples_leaf=2,
min_samples_split=5, n_estimators=200, n_jobs=-1, oob_score=True,
random_state=42)
```

Initialize Random Forest with Entropy (Information Gain) criterion

```
rf_entropy = RandomForestClassifier(
    criterion='entropy',
    n_estimators=200,
    max_depth=15,
    min_samples_split=5,
    min_samples_leaf=2,
    oob_score=True,
    random_state=42,
    n_jobs=-1
)

print("Entropy model initialized")
print(rf_entropy)
```

Output:

```
Entropy model initialized
RandomForestClassifier(criterion='entropy', max_depth=15, min_samples_leaf=2,
min_samples_split=5, n_estimators=200, n_jobs=-1, oob_score=True,
random_state=42)
```

Train both models on the same training data for fair comparison

```
rf_gini.fit(X_train, y_train)
print(f"Gini model trained -- OOB Score: {rf_gini.oob_score_:.4f}")

rf_entropy.fit(X_train, y_train)
print(f"Entropy model trained -- OOB Score: {rf_entropy.oob_score_:.4f}")
```

Output:
```
Gini model trained -- OOB Score: 0.9160
Entropy model trained -- OOB Score: 0.9148
```

Step 6: Model Evaluation

Generate predictions from both models on the same test set

```
y_pred_gini = rf_gini.predict(X_test)
y_pred_entropy = rf_entropy.predict(X_test)

print(f"Predictions generated for {len(y_pred_gini)} test samples (both models)")
```

Output:
```
Predictions generated for 8238 test samples (both models)
```

Compare accuracy and OOB scores in a comparison table

```
acc_gini = accuracy_score(y_test, y_pred_gini)
acc_entropy = accuracy_score(y_test, y_pred_entropy)

print("=" * 55)
print("       COMPARISON: Gini vs Entropy")
print("=" * 55)
print(f"  {'Metric':10} {'Entropy':>10}")
print("-" * 55)
print(f"  {'Test Accuracy':10.4f} {acc_entropy:>10.4f}")
print(f"  {'OOB Score':10.4f} {rf_entropy.oob_score_:>10.4f}")
print(f"  {'Difference (Gini - Entropy)':10.4f} {rf_gini.oob_score_ -
    rf_entropy.oob_score_:>10.4f}")
print("=" * 55)
```

Output:
```
=======================================================
COMPARISON: Gini vs Entropy
=======================================================
Metric                              Gini    Entropy
-------------------------------------------------------
Test Accuracy                     0.9226     0.9226
OOB Score                         0.9160     0.9148
Difference (Gini - Entropy)       0.0000     0.0012
=======================================================
```

Out-of-Bag Score explanation -- why OOB matters

```
print("Out-of-Bag (OOB) Estimation")
print("-" * 45)
print("Each tree in a Random Forest is trained on a")
print("bootstrap sample (~63% of data). The remaining")
print("~37% are 'out-of-bag' samples used to estimate")
print("generalization error WITHOUT a validation set.")
print()
print(f"Gini OOB Score:    {rf_gini.oob_score_:.4f}")
print(f"Entropy OOB Score: {rf_entropy.oob_score_:.4f}")
```

Output:

```
Out-of-Bag (OOB) Estimation
--------------------------------------------
Each tree in a Random Forest is trained on a
bootstrap sample (~63% of data). The remaining
~37% are 'out-of-bag' samples used to estimate
generalization error WITHOUT a validation set.
Gini OOB Score:    0.9160
Entropy OOB Score: 0.9148
```

Classification report -- Gini Impurity

```
print("Classification Report -- Gini Impurity")
print("-" * 55)
print(classification_report(y_test, y_pred_gini))
```

Output:

```
Classification Report -- Gini Impurity
-------------------------------------------------------
precision    recall  f1-score   support
0       0.94      0.97      0.96      7310
1       0.71      0.53      0.61       928
accuracy                           0.92      8238
macro avg       0.83      0.75      0.78      8238
weighted avg       0.92      0.92      0.92      8238
```

Classification report -- Entropy (Information Gain)

```
print("Classification Report -- Entropy")
print("-" * 55)
print(classification_report(y_test, y_pred_entropy))
```

Output:

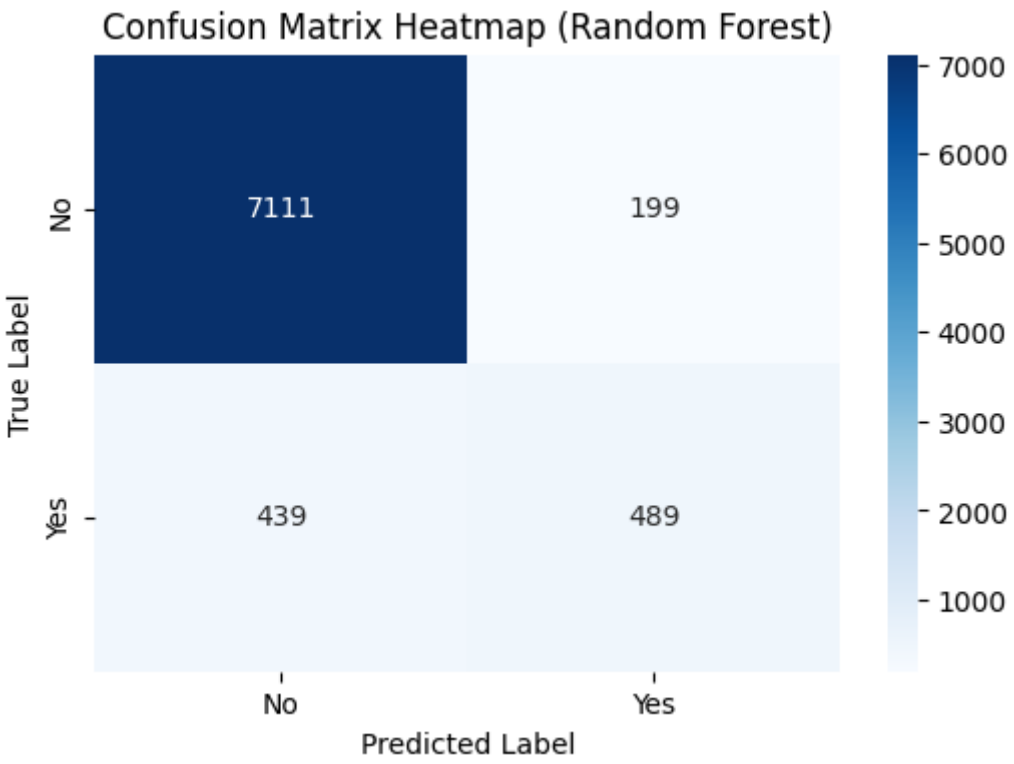
```
Classification Report -- Entropy
-------------------------------------------------------
precision    recall  f1-score   support
0       0.94      0.98      0.96      7310
1       0.72      0.51      0.60       928
accuracy                           0.92      8238
macro avg       0.83      0.74      0.78      8238
weighted avg       0.92      0.92      0.92      8238
```

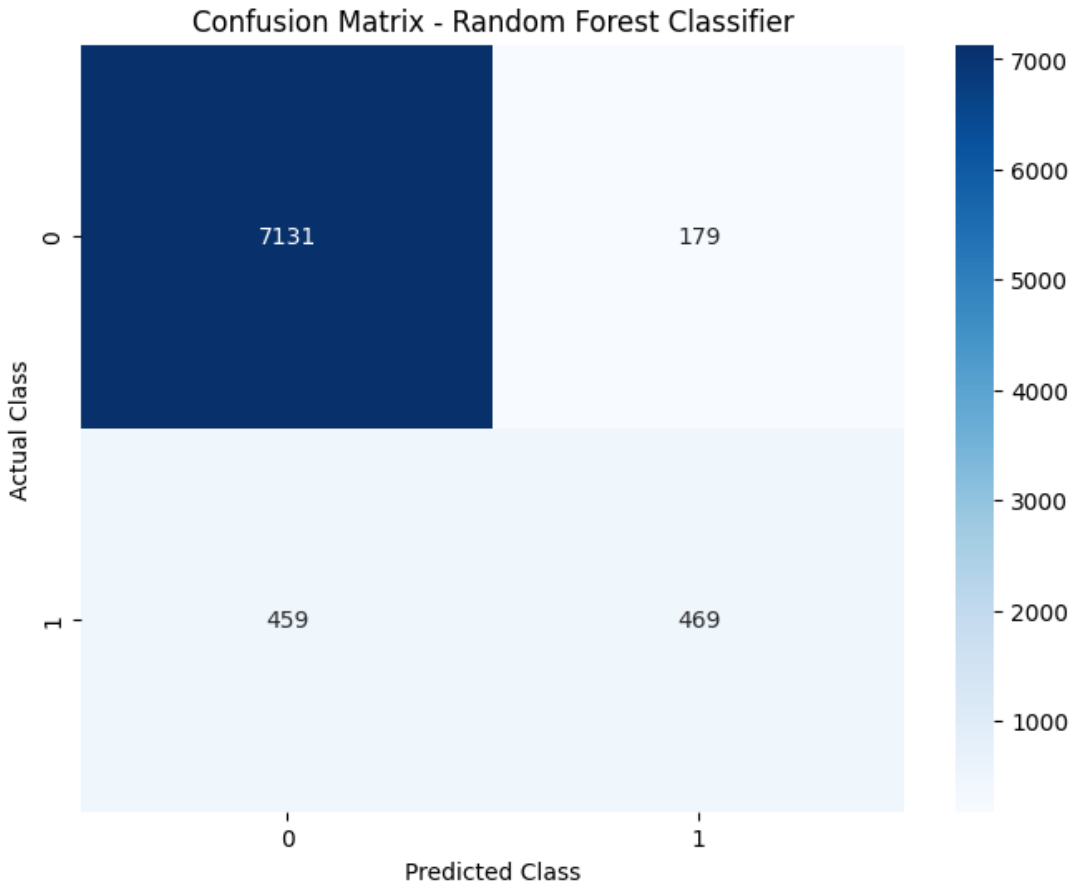
Display confusion matrices side-by-side for visual comparison

```
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

cm_gini = confusion_matrix(y_test, y_pred_gini)
sns.heatmap(cm_gini, annot=True, fmt='d', cmap='Blues',
            xticklabels=['No','Yes'], yticklabels=['No','Yes'], ax=axes[0])
axes[0].set_title("Gini Impurity")
axes[0].set_xlabel("Predicted"); axes[0].set_ylabel("Actual")

cm_entropy = confusion_matrix(y_test, y_pred_entropy)
sns.heatmap(cm_entropy, annot=True, fmt='d', cmap='Oranges',
            xticklabels=['No','Yes'], yticklabels=['No','Yes'], ax=axes[1])
axes[1].set_title("Entropy")
axes[1].set_xlabel("Predicted"); axes[1].set_ylabel("Actual")

plt.suptitle("Confusion Matrix Comparison", fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()
```

Output:

```
Gini Impurity
Entropy
```





Step 7: Model Simulation

Compare predictions from both models on a new sample

```
sample_data = [[56, 1, 1, 0, 0, 0, 0, 1, 0, 0, 261, 1, 999, 0, 0, 1.1, 93.994, -36.4,
    4.857, 5191.0]]

pred_gini = rf_gini.predict(sample_data)
pred_entropy = rf_entropy.predict(sample_data)
prob_gini = rf_gini.predict_proba(sample_data)[0]
prob_entropy = rf_entropy.predict_proba(sample_data)[0]

labels = {0: "No (will NOT subscribe)", 1: "Yes (will subscribe)"}

print("=" * 55)
print("  Sample Prediction Comparison")
print("=" * 55)
print(f"  Gini model    -> {labels[pred_gini[0]]}")
print(f"    Confidence  -> No: {prob_gini[0]:.3f}, Yes: {prob_gini[1]:.3f}")
print()
print(f"  Entropy model -> {labels[pred_entropy[0]]}")
print(f"    Confidence  -> No: {prob_entropy[0]:.3f}, Yes: {prob_entropy[1]:.3f}")
print("-" * 55)
if pred_gini[0] == pred_entropy[0]:
    print("  [ok] Both models agree on this prediction.")
else:
    print("  [x] Models disagree -- criterion choice matters here.")
print("=" * 55)
```

Output:

```
=======================================================
Sample Prediction Comparison
=======================================================
Gini model    -> No (will NOT subscribe)
Confidence  -> No: 0.997, Yes: 0.003
Entropy model -> No (will NOT subscribe)
Confidence  -> No: 0.995, Yes: 0.005
-------------------------------------------------------
[ok] Both models agree on this prediction.
=======================================================
```

Option B: Multiclass Classification (Seeds Dataset)

Step 1: Importing Libraries

Import core data handling libraries

```
import pandas as pd
import numpy as np

print("Pandas and Numpy imported")
```

Output:

```
Pandas and Numpy imported
```

Import machine learning models and utilities

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

print("Scikit-learn modules imported")
```

Output:

```
Scikit-learn modules imported
```

Import visualization tools

```
import matplotlib.pyplot as plt
import seaborn as sns

print("Visualization libraries imported")
```

Output:

```
Visualization libraries imported
```

Step 2: Loading Dataset

Define dataset column names

```
column_names = [
    "Area", "Perimeter", "Compactness",
    "Length_of_kernel", "Width_of_kernel",
    "Asymmetry_coefficient", "Length_of_kernel_groove", "Class"
]

print("Column names defined:", column_names)
```

Output:

```
Column names defined: ["Area", "Perimeter", "Compactness", "Length_of_kernel",
"Width_of_kernel", "Asymmetry_coefficient", "Length_of_kernel_groove", "Class"]
```

Load Seeds dataset from local file

```
df = pd.read_csv("seeds_dataset.txt", sep=r"\\s+", header=None, names=column_names)

print("Dataset loaded from seeds_dataset.txt")
```

Output:

```
Dataset loaded from seeds_dataset.txt
```

Step 3: Data Analysis

Check dataset dimensions

```
print("Dataset shape:", df.shape)
```

Output:

```
Dataset shape: (210, 8)
```

Preview first 5 records of the dataset

```
print("First 5 rows:")
print(df.head())
```

Output:

```
First 5 rows:
```

Output:

	Area	Perimeter	Compactness	Length_of_kernel	Width_of_kernel	Asymmetry_coefficient	Length_of_kernel_groove	Class
0	15.26	14.84	0.8710	5.763	3.312	2.221	5.220	1
1	14.88	14.57	0.8811	5.554	3.333	1.018	4.956	1
2	14.29	14.09	0.9050	5.291	3.337	2.699	4.825	1
3	13.84	13.94	0.8955	5.324	3.379	2.259	4.805	1
4	16.14	14.99	0.9034	5.658	3.562	1.355	5.175	1

Check the distribution of target classes

```
print("Class distribution:")
print(df["Class"].value_counts())
```

Output:

```
Class distribution:
1    70
2    70
3    70
Name: Class, dtype: int64
```

Generate descriptive statistical summary

```
print("Statistical summary:")
print(df.describe())
```

Output:

```
Statistical summary:
```

Output:

	Area	Perimeter	Compactness	Length_of_kernel	Width_of_kernel	Asymmetry_coefficient	Length_of_kernel_groove	Class
count	210.000000	210.000000	210.000000	210.000000	210.000000	210.000000	210.000000	210.000000
mean	14.847524	14.559286	0.870999	5.628533	3.258605	3.700201	5.408071	2.000000
std	2.909699	1.305959	0.023629	0.443063	0.377714	1.503557	0.491480	0.818448
min	10.590000	12.410000	0.808100	4.899000	2.630000	0.765100	4.519000	1.000000
25%	12.270000	13.450000	0.856900	5.262250	2.944000	2.561500	5.045000	1.000000
50%	14.355000	14.320000	0.873450	5.523500	3.237000	3.599000	5.223000	2.000000
75%	17.305000	15.715000	0.887775	5.979750	3.561750	4.768750	5.877000	3.000000
max	21.180000	17.250000	0.918300	6.675000	4.033000	8.456000	6.550000	3.000000

Step 4: Data Preprocessing

Separate features and target variable

```
X = df.drop("Class", axis=1)
y = df["Class"]

print("Features and target separated")
```

Output:

```
Features and target separated
```

Split data into training (75%) and testing (25%) sets

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42, stratify=y
)

print(f"Training set: {len(X_train)} samples")
print(f"Testing set: {len(X_test)} samples")
```

Output:
```
Training set: 157 samples
Testing set: 53 samples
```

Option B-i: Multiclass - Gini Impurity

Step 5: Model Training

Initialize Random Forest with GINI criterion

```
rf_gini = RandomForestClassifier(n_estimators=100, criterion="gini", oob_score=True,
    random_state=42)

print("Random Forest model (Gini) initialized with OOB scoring")
```

Output:
```
Random Forest model (Gini) initialized with OOB scoring
```

Train Random Forest model using GINI

```
rf_gini.fit(X_train, y_train)

print("Model training completed")
print(rf_gini)
```

Output:
```
Model training completed
RandomForestClassifier(oob_score=True, random_state=42)
```

Step 6: Model Evaluation

Calculate and display GINI model accuracy

```
accuracy = accuracy_score(y_test, rf_gini.predict(X_test))

print(f"Model accuracy: {accuracy}")
```

Output:
```
Model accuracy: 0.9056603773584906
```

Out-of-Bag (OOB) Score -- generalization estimate without a separate validation set

```
print(f"OOB Score: {rf_gini.oob_score_:.4f}")
print()
print("Each tree is evaluated on the ~37% of training samples")
print("that were left out of its bootstrap sample.")
```

Output:

```
OOB Score: 0.9108
Each tree is evaluated on the ~37% of training samples
that were left out of its bootstrap sample.
```

Display detailed classification report for GINI model

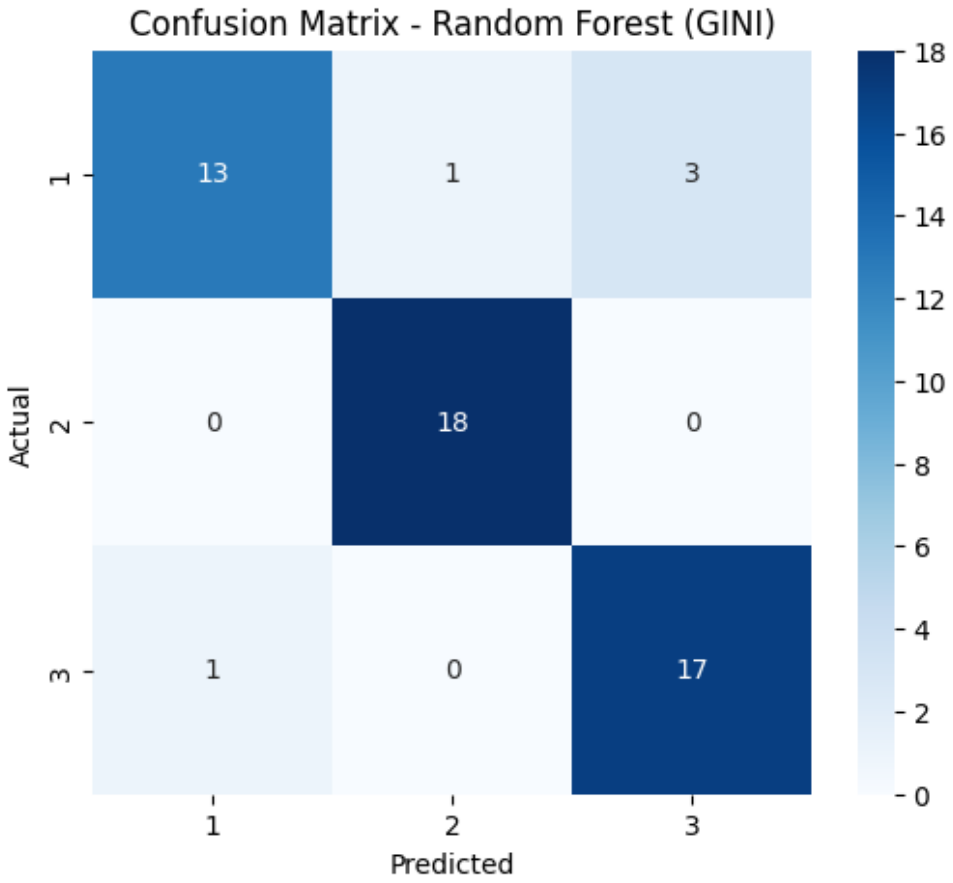
```
print(classification_report(y_test, rf_gini.predict(X_test)))
```

Output:

```
precision    recall  f1-score   support
1       0.93      0.76      0.84        17
2       0.95      1.00      0.97        18
3       0.85      0.94      0.89        18
accuracy                           0.91        53
macro avg       0.91      0.90      0.90        53
weighted avg       0.91      0.91      0.90        53
```

Plot confusion matrix for GINI model predictions

```
cm_gini = confusion_matrix(y_test, rf_gini.predict(X_test))
sns.heatmap(cm_gini, annot=True, fmt="d", cmap="Blues")
plt.title("Confusion Matrix - GINI")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



Step 7: Model Simulation

Select sample test indices representing different classes

```
sample_idx = [0, 20, 40]

print("Sample indices selected:", sample_idx)
```

Output:
```
Sample indices selected: [0, 20, 40]
```

Perform manual prediction and compare with true labels

```
samples = X_test.iloc[sample_idx]
true_labels = y_test.iloc[sample_idx]
preds = rf_gini.predict(samples)

for i in range(len(sample_idx)):
    print(f"Sample {i+1} Prediction: {preds[i]} (True: {true_labels.iloc[i]})")
```

Output:
```
Sample 1 Prediction: 1 (True: 1)
Sample 2 Prediction: 2 (True: 2)
Sample 3 Prediction: 3 (True: 3)
```

Option B-ii: Multiclass - Entropy

Step 5: Model Training

Initialize Random Forest with ENTROPY criterion

```
rf_entropy = RandomForestClassifier(n_estimators=100, criterion="entropy", oob_score=True,
    random_state=42)

print("Random Forest model (Entropy) initialized with OOB scoring")
```

Output:
```
Random Forest model (Entropy) initialized with OOB scoring
```

Train Random Forest model using ENTROPY

```
rf_entropy.fit(X_train, y_train)

print("Model training completed")
print(rf_entropy)
```

Output:
```
Model training completed
RandomForestClassifier(criterion='entropy', oob_score=True, random_state=42)
```

Step 6: Model Evaluation

Calculate and display ENTROPY model accuracy

```
accuracy = accuracy_score(y_test, rf_entropy.predict(X_test))

print(f"Model accuracy: {accuracy}")
```

Output:

```
Model accuracy: 0.8867924528301887
```

Out-of-Bag (OOB) Score -- generalization estimate without a separate validation set

```
print(f"OOB Score: {rf_entropy.oob_score_:.4f}")
print()
print("Each tree is evaluated on the ~37% of training samples")
print("that were left out of its bootstrap sample.")
```

Output:

```
OOB Score: 0.8981
Each tree is evaluated on the ~37% of training samples
that were left out of its bootstrap sample.
```

Display detailed classification report for ENTROPY model

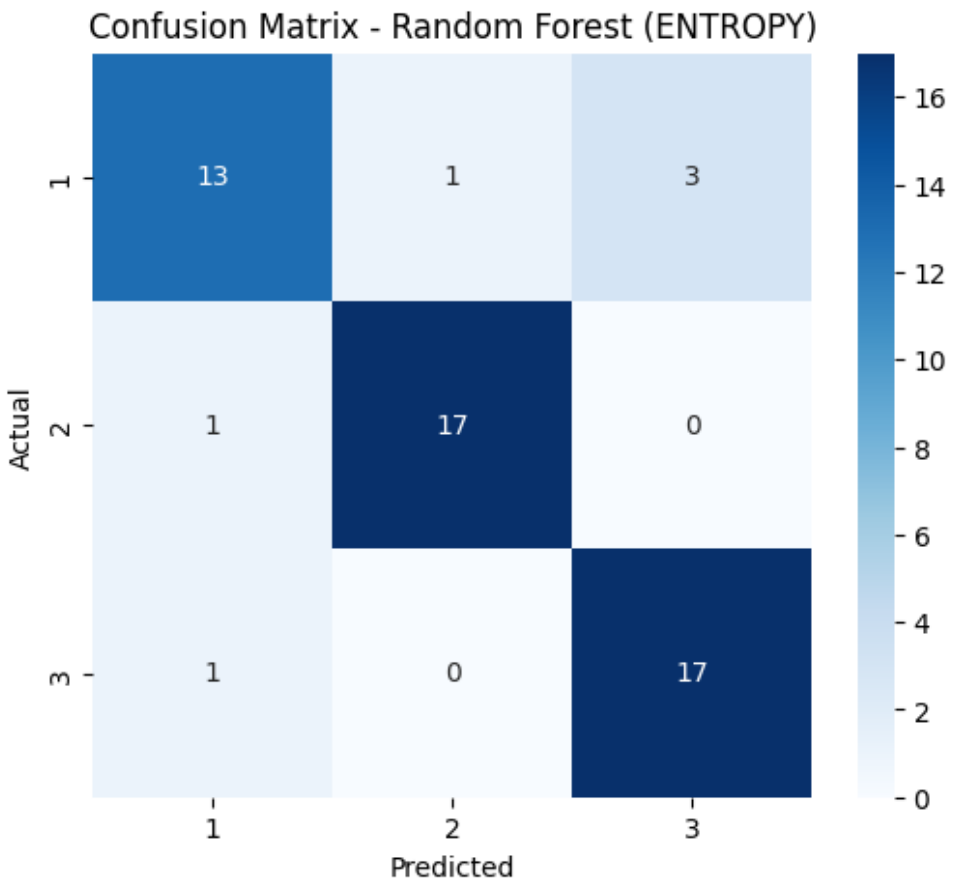
```
print(classification_report(y_test, rf_entropy.predict(X_test)))
```

Output:

```
Classification Report:
precision    recall  f1-score   support
1       0.88      0.82      0.85        17
2       0.95      1.00      0.97        18
3       0.84      0.83      0.83        18
accuracy                           0.89        53
macro avg       0.89      0.89      0.88        53
weighted avg       0.89      0.89      0.89        53
```

Plot confusion matrix for ENTROPY model predictions

```
cm_entropy = confusion_matrix(y_test, rf_entropy.predict(X_test))
sns.heatmap(cm_entropy, annot=True, fmt="d", cmap="Blues")
plt.title("Confusion Matrix - ENTROPY")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



Step 7: Model Simulation

Select sample test indices representing different classes

```
sample_idx = [0, 20, 40]

print("Sample indices selected:", sample_idx)
```

Output:

```
Sample indices selected: [0, 20, 40]
```

Perform manual prediction and compare with true labels

```
samples = X_test.iloc[sample_idx]
true_labels = y_test.iloc[sample_idx]
preds = rf_entropy.predict(samples)

for i in range(len(sample_idx)):
    print(f"Sample {i+1} Prediction: {preds[i]} (True: {true_labels.iloc[i]})")
```

Output:

```
Sample 1 Prediction: 1 (True: 1)
Sample 2 Prediction: 2 (True: 2)
Sample 3 Prediction: 3 (True: 3)
```

Option B-iii: Multiclass - Both (Gini vs Entropy)

Step 5: Model Training

Initialize Random Forest with Gini impurity criterion

```
rf_gini = RandomForestClassifier(
    n_estimators=100,
    criterion="gini",
    max_depth=None,
    min_samples_split=2,
    oob_score=True,
    random_state=42,
    n_jobs=-1
)

print("Gini model initialized")
print(rf_gini)
```

Output:

```
Gini model initialized
RandomForestClassifier(n_estimators=100, criterion='gini', oob_score=True,
random_state=42, n_jobs=-1)
```

Initialize Random Forest with Entropy (Information Gain) criterion

```
rf_entropy = RandomForestClassifier(
    n_estimators=100,
    criterion="entropy",
    max_depth=None,
    min_samples_split=2,
    oob_score=True,
    random_state=42,
    n_jobs=-1
)

print("Entropy model initialized")
print(rf_entropy)
```

Output:

```
Entropy model initialized
RandomForestClassifier(n_estimators=100, criterion='entropy', oob_score=True,
random_state=42, n_jobs=-1)
```

Train both models on the same training data for fair comparison

```
rf_gini.fit(X_train, y_train)
print(f"Gini model trained -- OOB Score: {rf_gini.oob_score_:.4f}")

rf_entropy.fit(X_train, y_train)
print(f"Entropy model trained -- OOB Score: {rf_entropy.oob_score_:.4f}")
```

Output:
```
Gini model trained -- OOB Score: 0.9108
Entropy model trained -- OOB Score: 0.8981
```

Step 6: Model Evaluation

Generate predictions from both models on the same test set

```
y_pred_gini = rf_gini.predict(X_test)
y_pred_entropy = rf_entropy.predict(X_test)

print(f"Predictions generated for {len(y_pred_gini)} test samples (both models)")
```

Output:
```
Predictions generated for 53 test samples (both models)
```

Compare accuracy and OOB scores in a comparison table

```
acc_gini = accuracy_score(y_test, y_pred_gini)
acc_entropy = accuracy_score(y_test, y_pred_entropy)

print("=" * 55)
print("       COMPARISON: Gini vs Entropy (Seeds)")
print("=" * 55)
print(f"  {'Metric':10} {'Entropy':>10}")
print("-" * 55)
print(f"  {'Test Accuracy':10.4f} {acc_entropy:>10.4f}")
print(f"  {'OOB Score':10.4f} {rf_entropy.oob_score_:>10.4f}")
print(f"  {'Difference (Gini - Entropy)':10.4f} {rf_gini.oob_score_ -
    rf_entropy.oob_score_:>10.4f}")
print("=" * 55)
```

Output:
```
=======================================================
COMPARISON: Gini vs Entropy (Seeds)
=======================================================
Metric                              Gini    Entropy
-------------------------------------------------------
Test Accuracy                     0.9057     0.8868
OOB Score                         0.9108     0.8981
Difference (Gini - Entropy)       0.0189     0.0127
=======================================================
```

Out-of-Bag Score explanation -- why OOB matters

```
print("Out-of-Bag (OOB) Estimation")
print("-" * 45)
print("Each tree in a Random Forest is trained on a")
print("bootstrap sample (~63% of data). The remaining")
print("~37% are 'out-of-bag' samples used to estimate")
print("generalization error WITHOUT a validation set.")
print()
print(f"Gini OOB Score:    {rf_gini.oob_score_:.4f}")
print(f"Entropy OOB Score: {rf_entropy.oob_score_:.4f}")
```

Output:

```
Out-of-Bag (OOB) Estimation
--------------------------------------------
Each tree in a Random Forest is trained on a
bootstrap sample (~63% of data). The remaining
~37% are 'out-of-bag' samples used to estimate
generalization error WITHOUT a validation set.
Gini OOB Score:    0.9108
Entropy OOB Score: 0.8981
```

Classification report -- Gini Impurity

```
print("Classification Report -- Gini Impurity")
print("-" * 55)
print(classification_report(y_test, rf_gini.predict(X_test)))
```

Output:

```
Classification Report -- Gini Impurity
-------------------------------------------------------
precision    recall  f1-score   support
1       0.93      0.76      0.84        17
2       0.95      1.00      0.97        18
3       0.85      0.94      0.89        18
accuracy                           0.91        53
macro avg       0.91      0.90      0.90        53
weighted avg       0.91      0.91      0.90        53
```

Classification report -- Entropy (Information Gain)

```
print("Classification Report -- Entropy")
print("-" * 55)
print(classification_report(y_test, rf_entropy.predict(X_test)))
```

Output:

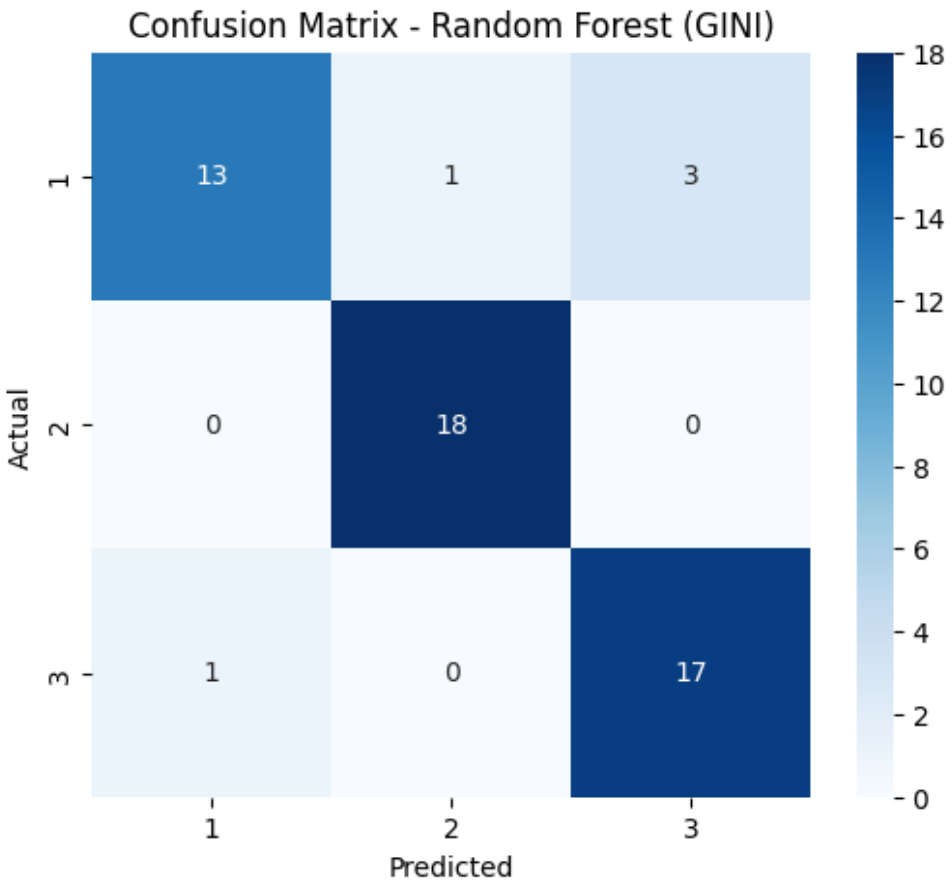
```
Classification Report -- Entropy
-------------------------------------------------------
precision    recall  f1-score   support
1       0.88      0.82      0.85        17
2       0.95      1.00      0.97        18
3       0.84      0.83      0.83        18
accuracy                           0.89        53
macro avg       0.89      0.89      0.88        53
weighted avg       0.89      0.89      0.89        53
```

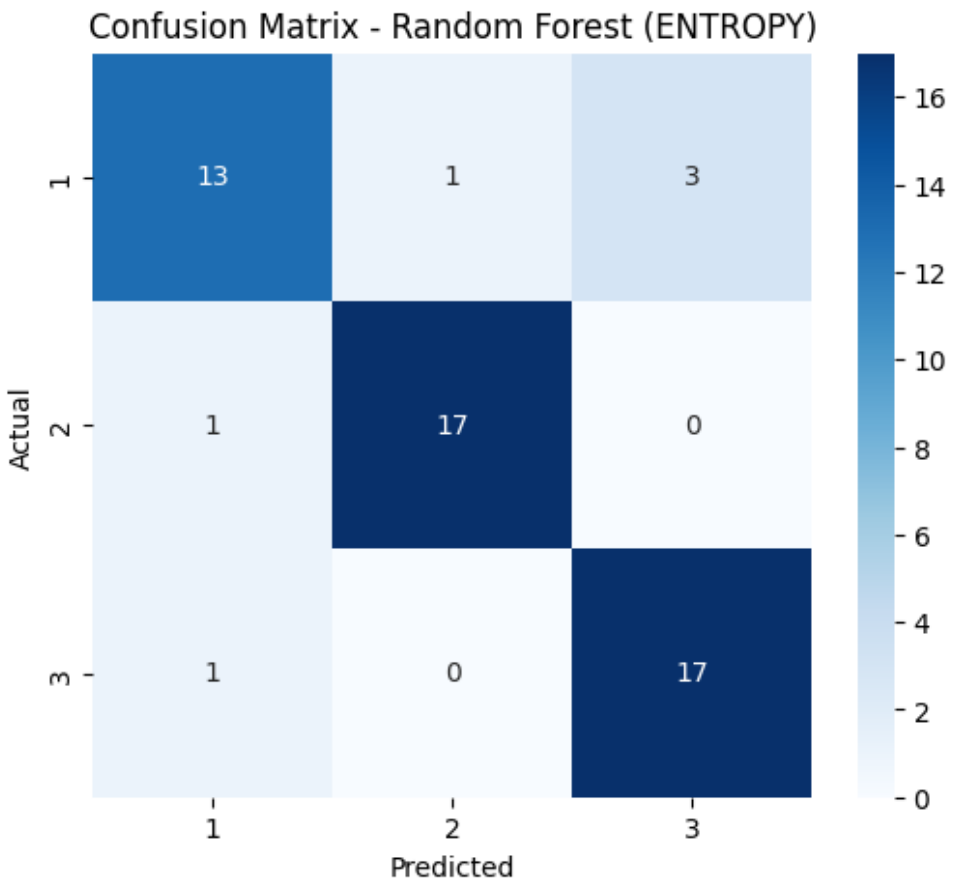
Display confusion matrices side-by-side for visual comparison

```
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

cm_gini = confusion_matrix(y_test, rf_gini.predict(X_test))
sns.heatmap(cm_gini, annot=True, fmt="d", cmap="Blues",
            xticklabels=[1,2,3], yticklabels=[1,2,3], ax=axes[0])
axes[0].set_title("Gini Impurity")
axes[0].set_xlabel("Predicted"); axes[0].set_ylabel("Actual")

cm_entropy = confusion_matrix(y_test, rf_entropy.predict(X_test))
sns.heatmap(cm_entropy, annot=True, fmt="d", cmap="Oranges",
            xticklabels=[1,2,3], yticklabels=[1,2,3], ax=axes[1])
axes[1].set_title("Entropy")
axes[1].set_xlabel("Predicted"); axes[1].set_ylabel("Actual")

plt.suptitle("Confusion Matrix Comparison", fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()
```

Output:

```
Gini Impurity
Entropy
```





Step 7: Model Simulation

Compare predictions from both models on multiple samples

```
sample_idx = [0, 20, 40]
samples = X_test.iloc[sample_idx]
true_labels = y_test.iloc[sample_idx]

preds_gini = rf_gini.predict(samples)
preds_entropy = rf_entropy.predict(samples)
probs_gini = rf_gini.predict_proba(samples)
probs_entropy = rf_entropy.predict_proba(samples)

label_map = {1: "Canadian", 2: "Kama", 3: "Rosa"}

print("=" * 60)
print("  Sample Prediction Comparison -- Gini vs Entropy")
print("=" * 60)
print(f"  {'Sample':<10} {'True':<12} {'Gini':<12} {'Entropy':<12} {'Match?'}")
print("-" * 60)
for i in range(len(sample_idx)):
    match = "[ok]" if preds_gini[i] == preds_entropy[i] else "[x]"
    true_name = label_map[true_labels.iloc[i]]
    gini_name = label_map[preds_gini[i]]
    entropy_name = label_map[preds_entropy[i]]
    print(f"  Sample {i+1:<4} {true_name:<12} {gini_name:<12} {entropy_name:<12} {match}")
print("-" * 60)

agree = sum(1 for g, e in zip(preds_gini, preds_entropy) if g == e)
print(f"  Agreement: {agree}/{len(sample_idx)} samples")
print("=" * 60)
```

Output:

```
============================================================
  Sample Prediction Comparison -- Gini vs Entropy
============================================================
  Sample     True         Gini         Entropy      Match?
------------------------------------------------------------
  Sample 1   Canadian     Canadian     Canadian     [ok]
  Sample 2   Kama         Kama         Kama         [ok]
  Sample 3   Rosa         Rosa         Rosa         [ok]
------------------------------------------------------------
  Agreement: 3/3 samples
============================================================
```